

# Polynomial Operations

## Practice Paper — Digital SAT Style

Difficulty Mix: Easy (Q1–10) | Medium (Q11–22) | Hard (Q23–30) · Total: 30 Questions

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### DIFFICULTY: EASY

**Q1.** What is  $(3x^2 + 5x - 2) + (x^2 - 3x + 7)$ ?

- A)  $4x^2 + 2x + 5$
  - B)  $4x^2 + 8x + 5$
  - C)  $4x^2 + 2x - 5$
  - D)  $2x^2 + 2x + 5$
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### DIFFICULTY: EASY

**Q2.** Simplify:  $(7x - 4) - (3x + 2)$

- A)  $4x - 6$
  - B)  $4x - 2$
  - C)  $4x + 6$
  - D)  $10x - 2$
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### DIFFICULTY: EASY

**Q3.** Which expression is equivalent to  $2x(3x - 5)$ ?

- A)  $6x - 10x$
  - B)  $6x^2 - 10$
  - C)  $6x^2 - 10x$
  - D)  $5x^2 - 10x$
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### DIFFICULTY: EASY

**Q4.** What is the degree of the polynomial  $4x^3 - 7x^2 + x - 9$ ?

- A) 1
  - B) 2
  - C) 3
  - D) 4
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**DIFFICULTY: EASY**

**Q5.** Which expression is equivalent to  $(x + 3)(x + 4)$ ?

- A)  $x^2 + 7x + 12$
  - B)  $x^2 + 12x + 7$
  - C)  $x^2 + 7x + 7$
  - D)  $x^2 + x + 12$
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**DIFFICULTY: EASY**

**Q6.** Simplify:  $3x^2(2x + 1) - x(x + 4)$

- A)  $6x^3 + 3x^2 - x^2 - 4x$
  - B)  $6x^3 + 2x^2 - 4x$
  - C)  $6x^3 + 3x^2 + x^2 + 4x$
  - D)  $5x^3 + 4x$
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**DIFFICULTY: EASY**

**Q7.** What is the coefficient of  $x^2$  in the product  $(2x + 1)(x^2 - 3x + 4)$ ?

- A) -6
  - B) -5
  - C) -4
  - D) 8
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**DIFFICULTY: EASY**

**Q8.** Which of the following is equal to  $(x - 5)^2$ ?

- A)  $x^2 - 25$
  - B)  $x^2 + 25$
  - C)  $x^2 - 10x + 25$
  - D)  $x^2 + 10x + 25$
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**DIFFICULTY: EASY**

**Q9.** What is the leading coefficient of  $-3x^4 + 2x^3 - x + 7$ ?

- A) 7
- B) 1
- C) -1

D) -3

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**DIFFICULTY: EASY**

**Q10.** If  $p(x) = x^2 - 4$  and  $q(x) = x + 2$ , what is  $p(x) - q(x)$ ?

A)  $x^2 - x - 6$

B)  $x^2 + x - 6$

C)  $x^2 - x - 2$

D)  $x^2 + x - 2$

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**DIFFICULTY: MEDIUM**

**Q11.** What is  $(x + 3)(x^2 - 3x + 9)$ ?

A)  $x^3 - 27$

B)  $x^3 + 27$

C)  $x^3 - 9$

D)  $x^3 + 9x$

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**DIFFICULTY: MEDIUM**

**Q12.** Which expression is equivalent to  $(2x - 3)^2$ ?

A)  $4x^2 - 9$

B)  $4x^2 + 9$

C)  $4x^2 - 12x + 9$

D)  $4x^2 + 12x + 9$

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**DIFFICULTY: MEDIUM**

**Q13.** What is the remainder when  $x^3 - 2x^2 + 3x - 5$  is divided by  $(x - 1)$ ?

A) -5

B) -3

C) 3

D) 5

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**DIFFICULTY: MEDIUM**

**Q14.** Which of the following is a factor of  $x^3 - 8$ ?

A)  $x + 2$

- B)  $x - 2$
  - C)  $x^2 + 2x - 4$
  - D)  $x^2 + 2x + 4$
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**DIFFICULTY: MEDIUM**

**Q15.** What is the coefficient of  $x^2$  in the expansion of  $(x + 2)(x - 1)(x + 3)$ ?

- A) 1
  - B) 2
  - C) 4
  - D) 8
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**DIFFICULTY: MEDIUM**

**Q16.** If  $f(x) = 2x^3 - x^2 + 4x - 3$ , what is  $f(-1)$ ?

- A) -10
  - B) -8
  - C) -6
  - D) -4
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**DIFFICULTY: MEDIUM**

**Q17.** Which polynomial when multiplied by  $(x - 2)$  gives  $x^3 - 8$ ?

- A)  $x^2 + 2x + 4$
  - B)  $x^2 - 2x + 4$
  - C)  $x^2 + 4$
  - D)  $x^2 - 4$
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**DIFFICULTY: MEDIUM**

**Q18.** What is  $(3x^2 - 2x + 1)(x + 4)$ ?

- A)  $3x^3 + 10x^2 - 7x + 4$
  - B)  $3x^3 + 10x^2 + 7x - 4$
  - C)  $3x^3 - 10x^2 - 7x + 4$
  - D)  $3x^3 + 10x^2 - 7x - 4$
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**DIFFICULTY: MEDIUM**

**Q19.** When the polynomial  $p(x) = x^3 + ax^2 + bx + 6$  is divided by  $(x - 1)$ , the remainder is 4.

When divided by  $(x + 1)$ , the remainder is -2. What is the value of  $a$ ?

- A) -5
  - B) -3
  - C) 1
  - D) 3
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**DIFFICULTY: MEDIUM**

**Q20.** What is the sum of the coefficients of  $(x - 2)^3$ ?

- A) -7
  - B) -1
  - C) 1
  - D) 7
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**DIFFICULTY: MEDIUM**

**Q21.** If  $(x^2 - 1)(x + k) = x^3 + kx^2 - x - k$ , which value of  $k$  makes this a valid identity?

- A) Any value of  $k$
  - B)  $k = 1$  only
  - C)  $k = -1$  only
  - D)  $k = 0$  only
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**DIFFICULTY: MEDIUM**

**Q22.** The polynomial  $p(x) = x^4 - 16$  can be written as:

- A)  $(x^2 - 4)^2$
  - B)  $(x^2 - 4)(x^2 + 4)$
  - C)  $(x - 2)(x + 2)(x^2 - 4)$
  - D)  $(x - 4)(x + 4)$
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**DIFFICULTY: HARD**

**Q23.** If  $(x^2 + bx + c)(x - 3) = x^3 - x^2 - 9x + 9$ , what is  $b + c$ ?

- A) -1
- B) 0
- C) 2

D) 5

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**DIFFICULTY: HARD**

**Q24.** When  $x^3 - 5x^2 + 4$  is divided by  $(x - 1)$ , what is the quotient?

- A)  $x^3 + x^2 - 4x - 4$
  - B)  $x^3 - x^2 - 4x + 4$
  - C)  $x^3 + x^2 + 4x + 4$
  - D)  $x^3 - x^2 + 4x - 4$
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**DIFFICULTY: HARD**

**Q25.** If  $p(x) = x^3 - 3x^2 + 4$  and  $p(a) = 0$ , which of the following must also equal zero?

- A)  $p(-a)$
  - B)  $p(2a)$
  - C)  $p(a + 1)$
  - D)  $(x - a)$  is a factor of  $p(x)$
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**DIFFICULTY: HARD**

**Q26.** What is the coefficient of  $x^2$  in the expansion of  $(x + 1)^5$ ?

- A) 5
  - B) 10
  - C) 15
  - D) 20
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**DIFFICULTY: HARD**

**Q27.** The polynomial  $x^3 + px^2 + qx + 6$  has roots  $r$ ,  $s$ , and  $t$ .

If  $rst = -6$  and  $r + s + t = -p$ , which expression equals  $q$ ?

- A)  $rs + st + tr$
  - B)  $r + s + t$
  - C)  $-(rs + st + tr)$
  - D)  $rst$
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**DIFFICULTY: HARD**

**Q28.** If  $f(x) = (x^2 - 4) \cdot g(x) + (3x - 2)$  for all  $x$ , what is  $f(2)$ ?

- A) 2
  - B) 4
  - C) 6
  - D) 8
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**DIFFICULTY: HARD**

**Q29.** For which value of  $k$  is  $(x - 2)$  a factor of  $x^3 + kx^2 - 4x + 8$ ?

- A) -3
  - B) -2
  - C) 1
  - D) 2
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**DIFFICULTY: HARD**

**Q30.** If the product of three consecutive integers is 210, and the middle integer is  $n$ , which polynomial equation models this situation?

- A)  $n^3 - n = 210$
  - B)  $n^3 + n = 210$
  - C)  $n^3 - 3n + 2 = 210$
  - D)  $3n + 2 = 210$
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